Restaurant Table Reservation and Food Service Management System

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Course: Database Management System

Review 1

Problem Statement:

The existing restaurant management process often lacks a centralized system to coordinate table reservations, customer orders, kitchen processing, billing, discounts, and payments, which can result in reservation conflicts, order errors, service delays, inconsistent records, and difficulty in monitoring restaurant performance



Existing System Problems:

Problem	Impact
Reservation conflicts	Customer dissatisfaction
No real-time table status	Poor table utilization
Order miscommunication	Wrong/delayed orders
Kitchen status not visible	Slow service
Disconnected records	Data inconsistency
Billing/discount errors	Revenue loss
Poor reporting	Weak decision-making

Proposed solution:

The proposed system will provide a centralized database for managing restaurant operations.

Main Functions :

1) Manage dining areas and tables
2) Register customer information
3) Reserve tables
4) Manage menu items
5) Record customer orders
6) Track order items
7) Monitor kitchen status
8) Generate bills
9) Apply authorized discounts
10) Record payments

Project Objectives :



The main objectives are:

1) To avoid overlapping table reservations.

2) To maintain customer and reservation records.

3) To manage restaurant tables and dining areas.

4) To record customer food orders accurately.

5) To track food preparation through kitchen tickets.

6) To manage billing and discounts.

7) To record payment information.

8) To maintain data consistency and integrity.

9) To generate useful reports for restaurant management

Scope of the Project :

The system covers the following modules:

🪑 Table Management
Manage dining areas, tables, capacity and availability.

👤 Customer Management
Store customer details and reservation information.

📅 Reservation Management
Reserve tables and prevent booking conflicts.

🍔 Food Order Management
Manage menu items, orders and order items

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👨‍🍳 Kitchen Management
Track food preparation using kitchen tickets.

💰 Billing Management Generate bills and manage discounts.

💳 Payment Management
Record payment details and payment status.



Users and Their Roles :

Customer

1) Makes reservations.
2) Provides guest informatio
3) Places food orders

Kitchen Staff

1) Receives kitchen orders.
2) Updates preparation status.
3) Marks orders as ready.

Admin / Manager

1) Manages restaurant data.
2) Monitors operations.
3) Generates reports.

Waiter

1) Manages assigned tables.
2) Records customer orders.
3) Updates serving status.

Cashier

1) Generates bills.
2) Applies authorized discounts.
3) Records payments.

Functional Requirements :

The system should be able to:

1) Add and manage dining areas.

2) Add and manage restaurant tables.

3) Store customer details.

4) Create and manage table reservations.

5) Manage menu items.

6) Record customer orders.

7) Store multiple items for each order.

8) Generate kitchen tickets.

9) Update kitchen preparation status.

10) Generate customer bills.

11) Apply authorized discounts.

12) Record payments.

13) Generate restaurant reports.



Major Entities for ER Diagram :

The project specification identifies these principal entities:

1) Dining Area

2) Table

3) Customer

4) Reservation

5) Menu Item

6) Waiter

7) Order

8) Order Item

9) Kitchen Ticket

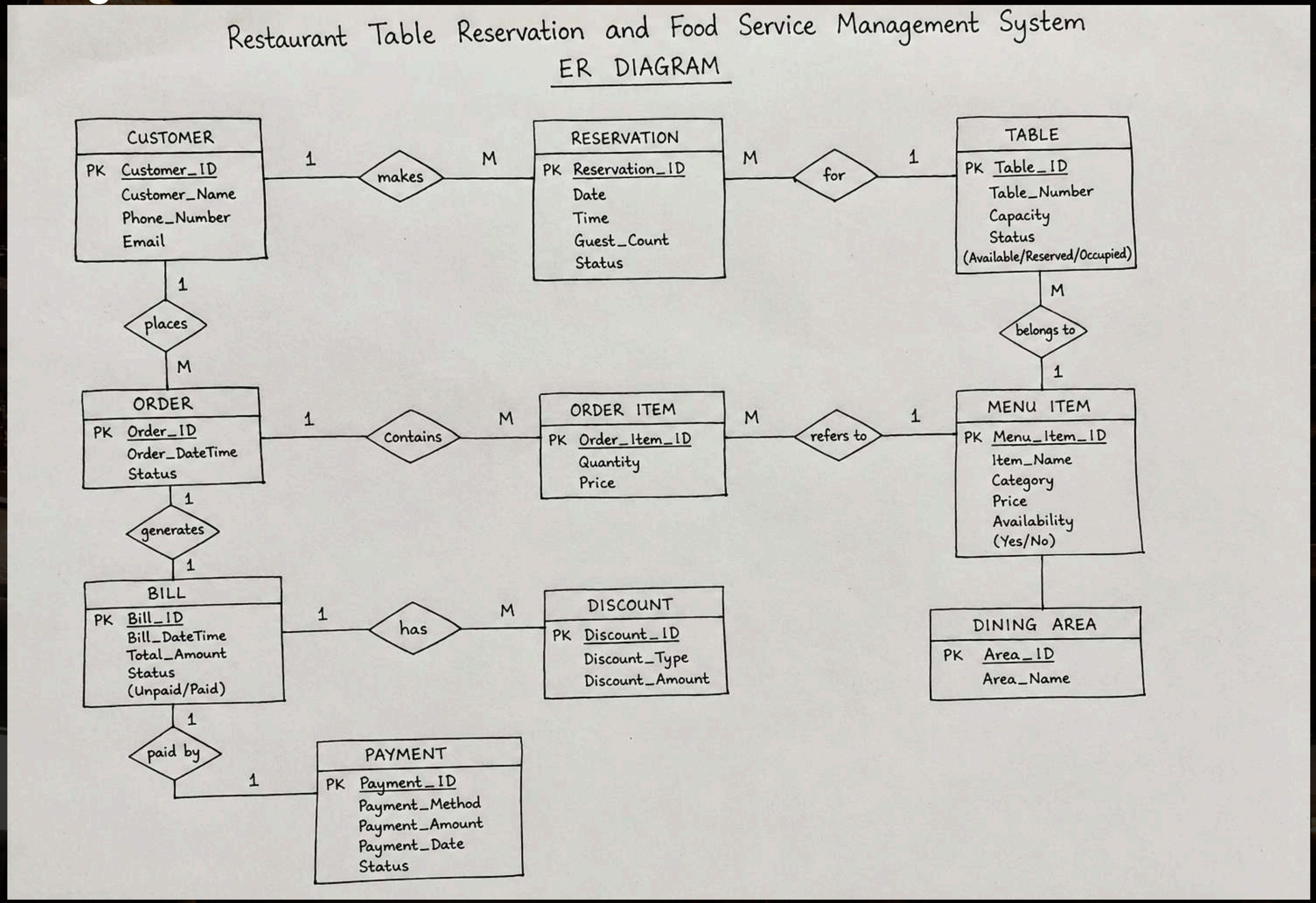
10) Bill

11) Discount

12) Payment



ER Diagram


Restaurant Table Reservation and Food Service Management System
ER DIAGRAM
CUSTOMER
PK Customer_ID
Customer_Name
Phone_Number
Email
1
makes
M
RESERVATION
PK Reservation_ID
Date
Time
Guest_Count
Status
M
for
1
TABLE
PK Table_ID
Table_Number
Capacity
Status
(Available/Reserved/Occupied)
1
places
M
M
belongs to
1
ORDER
PK Order_ID
Order_DateTime
Status
1
contains
M
ORDER ITEM
PK Order_Item_ID
Quantity
Price
M
refers to
1
MENU ITEM
PK Menu_Item_ID
Item_Name
Category
Price
Availability
(Yes/No)
1
generates
1
BILL
PK Bill_ID
Bill_DateTime
Total_Amount
Status
(Unpaid/Paid)
1
has
M
DISCOUNT
PK Discount_ID
Discount_Type
Discount_Amount
DINING AREA
PK Area_ID
Area_Name
1
paid by
1
PAYMENT
PK Payment_ID
Payment_Method
Payment_Amount
Payment_Date
Status

Initial Relational Schema :

Proposed Database Tables

Table Name	Primary Key (PK)	Important Foreign Key(s) (FK)
DINING_AREA	Area_ID	—
RESTAURANT_TABLE	Table_ID	Area_ID → DINING_AREA(Area_ID)
CUSTOMER	Customer_ID	—
WAITER	Waiter_ID	—
MENU_ITEM	Menu_Item_ID	—
RESERVATION RESERVATION	Reservation_ID Reservation_ID	Customer_ID → CUSTOMER(Customer_ID) Table_ID → RESTAURANT_TABLE(Table_ID)
FOOD_ORDER FOOD_ORDER FOOD_ORDER	Order_ID Order_ID Order_ID	Customer_ID → CUSTOMER(Customer_ID) Table_ID → RESTAURANT_TABLE(Table_ID) Waiter_ID → WAITER(Waiter_ID)
ORDER_ITEM ORDER_ITEM	Order_Item_ID Order_Item_ID	Order_ID → FOOD_ORDER(Order_ID) Menu_Item_ID → MENU_ITEM(Menu_Item_ID)
BILL	Bill_ID	Order_ID → FOOD_ORDER(Order_ID)
KITCHEN_TICKET	Ticket_ID	Order_ID → FOOD_ORDER(Order_ID)
PAYMENT	Payment_ID	Bill_ID → BILL(Bill_ID)
DISCOUNT	Discount_ID	Bill_ID → BILL(Bill_ID)

conclusion:

1) The project provides a centralized solution for managing restaurant operations.

2) It integrates table reservations, customer details, food orders, kitchen processing, billing, discounts, and payments.

3) The ER Diagram and Initial Relational Schema provide a clear foundation for the database design.

4) The system helps maintain organized and connected data across differentrestaurant operations.

5) In the next review, the database will be implemented using MySQL with tables, constraints, and SQL queries.

6) In Review 3, the database will be integrated with a web application to create a practical restaurant management system.

Future Scope :

1) Develop a complete web application for the restaurant.

2) Connect the website with the MySQL database.

3) Add online table reservation.

4) Add digital menu and food ordering.

5) Implement real-time kitchen order tracking.

6) Add online payment functionality.

7) Generate sales and restaurant management reports.